THE HERITAGE SCHOOL

FIRST TERM EXAMINATION, 2026–27

PHYSICS PAPER 1 (THEORY) · CLASS 12 · PRACTICE PAPER 1

Maximum Marks: 70 **Time Allotted: Three Hours**

Instructions to Candidates

- You are allowed an additional fifteen minutes for only reading the question paper. You must NOT start writing during this time.
- There are **twenty questions** in this paper. **Answer all questions.**
- There are **four sections** in the paper: A, B, C and D. Internal choices have been provided in **two questions** each in Sections B, C and D.
- **Section A** consists of one question having fourteen sub-parts of **1 mark** each.
- While attempting the multiple choice sub-parts you are required to write only ONE option as the answer.
- **Section B** consists of seven questions of **2 marks** each.
- **Section C** consists of nine questions of **3 marks** each.
- **Section D** consists of three questions of **5 marks** each.
- The intended marks for questions or parts of questions are given in brackets [ ].
- All working, including rough work, should be done on the same sheet as, and adjacent to, the rest of the answer.
- A list of useful physical constants is given at the end of this paper.
- A simple scientific calculator without a programmable memory may be used for calculations.

Syllabus covered: Electrostatics; Current Electricity; Magnetic Effect of Current and Magnetism; Electromagnetic Induction and Alternating Current; Electromagnetic Waves; Optics (Ray Optics and Optical Instruments, and Wave Optics).

Not examined in this paper: Dual Nature of Radiation and Matter, Atoms and Nuclei, Electronic Devices. Polarisation is no longer in the Wave Optics syllabus and is not examined here either.

SECTION A – 14 MARKS

Question 1 [14 × 1]

In sub-parts (i) to (vii) choose the correct alternative. Answer sub-parts (viii) to (xiv) briefly.

(i) A point charge q is placed at the centre of a cube. The electric flux emerging through **one face** of the cube is:

(a) q/ε_0
(b) $q/6\varepsilon_0$
(c) $q/8\varepsilon_0$
(d) zero

(ii) A uniform metallic wire of resistance R is stretched until its length is twice the original. Its new resistance is:

(a) $R/4$
(b) $R/2$
(c) $2R$
(d) $4R$

(iii) Two long straight parallel wires $2a$ apart carry equal currents I in **opposite** directions. The magnetic flux density at the mid-point of the line joining them is:

(a) zero
(b) $\mu_0 I/2\pi a$
(c) $\mu_0 I/\pi a$
(d) $2\mu_0 I/\pi a$

(iv) An ideal capacitor is connected across an alternating voltage source. The current in the circuit:

(a) leads the applied voltage by $\pi/2$.
(b) lags behind the applied voltage by $\pi/2$.
(c) is in phase with the applied voltage.
(d) leads the applied voltage by π.

(v) A concave mirror of focal length 12 cm forms a **real** image three times the size of the object. The object distance is:

(a) 8 cm
(b) 16 cm
(c) 36 cm
(d) 48 cm

(vi) **Assertion:** when the whole of a Young's double slit apparatus is immersed in water, the fringe width decreases. **Reason:** the wavelength of light decreases when it enters water.

(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Assertion is false and Reason is true.

(vii) **Assertion:** the kinetic energy of a charged particle moving in a uniform magnetic field remains constant. **Reason:** the magnetic force on a moving charge is always perpendicular to its velocity.

(a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Assertion is false and Reason is true.

(viii) State Gauss' theorem in electrostatics.

(ix) Name the conservation principle on which Kirchhoff's loop rule is based.

(x) Define the current sensitivity of a moving coil galvanometer.

(xi) What is the power factor of an ideal inductor connected to an a.c. source?

(xii) Name the electromagnetic radiation used for sterilising surgical instruments.

(xiii) A convex lens of focal length 20 cm is placed in contact with a concave lens of focal length 25 cm. Calculate the power of the combination.

(xiv) What is the phase difference between any two points lying on the same wavefront?

SECTION B – 14 MARKS

Question 2 [2]

Three capacitors of capacitance 2 μF, 3 μF and 6 μF are joined in **series** across a 24 V battery. Calculate the equivalent capacitance and the charge on each capacitor.

OR

Define electric dipole moment and state its SI unit. For what orientation of a dipole in a uniform electric field is the torque on it maximum?

Question 3 [2]

A potentiometer wire 10 m long has a resistance of 40 Ω. It is connected in series with a 60 Ω resistor and a driver cell of emf 4 V of negligible internal resistance. Calculate the potential gradient along the wire.

Question 4 [2]

State any two differences between a diamagnetic and a ferromagnetic substance.

Question 5 [2]

A coil of 100 turns and area 5×10^{-3} m^2 is held with its plane perpendicular to a uniform magnetic field of 0·20 T. It is rotated through 90° in 0·10 s. Calculate the average emf induced in it.

OR

In a step-down transformer, explain why the current in the secondary coil is greater than that in the primary.

Question 6 **[2]**

(i) Name the electromagnetic wave whose wavelength is 3 cm, and state any one of its uses.

(ii) Which physical quantity is the same for all electromagnetic waves travelling in vacuum?

Question 7 **[2]**

(i) An object is placed at a distance $2f$ from a convex lens of focal length f. State the position, nature and size of the image.

(ii) Define the power of a lens and state its SI unit.

Question 8 **[2]**

In a Young's double slit experiment the fringe width is 0·20 mm when the screen is 1·0 m from the slits.

(i) What will it be if the screen is moved to 1·5 m?

(ii) State the effect on the fringe width of using a source of shorter wavelength.

SECTION C – 27 MARKS

Question 9 **[3]**

Using Gauss' theorem, derive an expression for the intensity of the electric field at a point near a uniformly charged infinite plane sheet of surface charge density σ.

Question 10 **[3]**

E_1 = 16 V in a branch of total resistance 2 Ω and E_2 = 15 V in a branch of total resistance 3 Ω drive a common 4 Ω arm. Using Kirchhoff's laws, calculate the current in the 4 Ω resistor.

OR

Obtain the balance condition for a Wheatstone bridge network. State clearly the assumption you make at the start.

Question 11 **[3]**

Using Biot–Savart's law, derive an expression for the magnetic flux density at the centre of a circular coil of N turns and radius R carrying a current I.

Question 12 **[3]**

A moving coil galvanometer of resistance 99 Ω gives a full-scale deflection for a current of 10 mA.

(i) How would you convert it into an ammeter of range 0–1 A?

(ii) Calculate the resistance of the ammeter so obtained.

Question 13 **[3]**

Derive an expression for the coefficient of self-inductance of a long air-cored solenoid of length l, area of cross-section A and N turns. Hence state any two factors on which it depends.

Question 14 **[3]**

Using Huygens' wave theory and a neat labelled diagram, prove Snell's law for the refraction of light.

OR

In a single-slit Fraunhofer diffraction experiment, monochromatic light of wavelength 600 nm falls normally on a slit of width 0·30 mm, with the screen 1·5 m away. Calculate (a) the angular width and (b) the linear width of the central maximum.

Question 15 **[3]**

For any prism, show that the refractive index of its material is $n = \sin[(A+\delta_m)/2] / \sin(A/2)$, where the terms have their usual meaning.

Question 16 **[3]**

A compound microscope has an objective of focal length 2·0 cm and an eyepiece of focal length 6·25 cm, the two being 15 cm apart. If the final image is formed at the least distance of distinct vision, calculate the magnifying power.

Question 17 **[3]**

In a Young's double slit experiment, state how the interference pattern is affected by each of the following, with a reason.

(i) The intensity of the light through one slit is reduced.

(ii) The monochromatic source is replaced by white light.

(iii) One of the two slits is covered.

SECTION D – 15 MARKS

Question 18 **[5]**

(a) Derive the relation between the object distance u, the image distance v, the radius of curvature R and the refractive indices n_1 (rarer) and n_2 (denser) for refraction at a single convex spherical surface, taking the object in the rarer medium and a real image in the denser medium. **[3]**

(b) A point object in air is 40 cm from a convex spherical refracting surface of radius of curvature 15 cm separating air from glass of refractive index 1·5. Find the position of the image. **[2]**

OR

(a) Draw a labelled ray diagram of an astronomical telescope in normal adjustment and derive an expression for its magnifying power. **[3]**

(b) The objective and eyepiece of an astronomical telescope have focal lengths 150 cm and 6 cm. Calculate its magnifying power and its length in normal adjustment. **[2]**

Question 19 **[5]**

(a) A choke coil of self-inductance 0·50 H and resistance 30 Ω is connected to a 200 V, 50 Hz a.c. supply. Calculate the rms current and the power factor. **[3]**

(b) State the condition for resonance in a series LCR circuit and write the expression for the resonant frequency. **[2]**

OR

(a) An ideal step-down transformer converts 2200 V to 220 V. Its primary has 4000 turns and it delivers an output power of 8·0 kW. Calculate the number of secondary turns and the primary current. **[3]**

(b) State any two causes of energy loss in a transformer and how each is minimised. **[2]**

Question 20 **[5]**

Read the passage and answer the questions that follow.

The flash-gun of a camera does not draw its energy directly from the battery. The battery slowly charges a capacitor through a large resistance, and the capacitor is then discharged through the flash tube in a few milliseconds. A 100 μF capacitor charged from a 300 V supply through a 1·0 MΩ resistor is typical. The same idea is used in a heart defibrillator. In both devices what matters is not only the energy stored but the very short time in which it can be delivered.

(i) Write an expression for the energy stored in a charged capacitor in terms of C and V.

(ii) Calculate the energy stored in the 100 μF capacitor at 300 V.

(iii) If the flash lasts 2·0 ms, calculate the average power delivered.

(iv) Calculate the initial charging current through the 1·0 MΩ resistor.

(v) State one reason why a capacitor, rather than the battery itself, supplies the flash.

USEFUL CONSTANTS AND RELATIONS

	Quantity	Symbol	Value
1.	Permittivity of vacuum	ε_0	$8{\cdot}85 \times 10^{-12}$ F m^{-1}
2.	Constant for Coulomb's law	$1/4\pi\varepsilon_0$	9×10^{9} N m^{2} C^{-2}
3.	Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ H m^{-1}
4.	Constant in magnetism	$\mu_0/4\pi$	1×10^{-7} H m^{-1}
5.	Charge of a proton = charge of an electron	e	$1{\cdot}6 \times 10^{-19}$ C
6.	Mass of an electron	m_e	$9{\cdot}1 \times 10^{-31}$ kg
7.	Mass of a proton	m_p	$1{\cdot}67 \times 10^{-27}$ kg
8.	Speed of light in vacuum	c	3×10^{8} m s^{-1}
9.	Acceleration due to gravity	g	$9{\cdot}8$ m s^{-2}
10.	Least distance of distinct vision	D	25 cm

ANSWER KEY & MARKING NOTES

Practice Paper 1. **Do not read this section before attempting the paper.** Every numerical answer was recomputed independently; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	**(b)** $q/6\varepsilon_0$. The whole flux q/ε_0 is shared equally by the six faces, by symmetry.
1 (ii)	**(d)** 4*R*. Stretching conserves the **volume**, so doubling l halves A and $R = \rho l/A$ rises four-fold ($R \propto l^2$ at constant volume).
1 (iii)	**(c)** $\mu_0 I/\pi a$. For **antiparallel** currents the two fields at the mid-point point the same way and add: $2 \times \mu_0 I/2\pi a$.
1 (iv)	**(a)** In a capacitor the current leads the voltage by $\pi/2$ (CIVIL).
1 (v)	**(b)** 16 cm. Real image means $m = -3$, so $v = 3u$; substituting in $1/v + 1/u = 1/f$ with $f = -12$ cm gives $u = -16$ cm.
1 (vi)	**(a)** $\beta = \lambda D/d$ and λ falls to λ/n in water, so β falls by the factor n. The reason is exactly the cause.
1 (vii)	**(a)** A force always perpendicular to the velocity does no work, so the speed and hence the kinetic energy cannot change.
1 (viii)	The total electric flux through any **closed** surface in vacuum equals $1/\varepsilon_0$ times the net charge enclosed by it: $\oint \mathbf{E}\cdot d\mathbf{A} = q_{enc}/\varepsilon_0$.
1 (ix)	Conservation of **energy**. (The junction rule rests on conservation of **charge**.)
1 (x)	The deflection produced per unit current, $\varphi/I = NAB/C$.
1 (xi)	**Zero** (cos 90° = 0), so the current is wattless and no power is consumed.
1 (xii)	**Ultraviolet** radiation.
1 (xiii)	$P = P_1 + P_2 = 1/0{\cdot}20 + (-1/0{\cdot}25) = 5 - 4 =$ **+1 D**, i.e. a converging combination of focal length 1 m.
1 (xiv)	**Zero** – a wavefront is by definition the locus of points vibrating in the same phase.

Section B

Q	Answer
2	$1/C_s = 1/2 + 1/3 + 1/6 = 1$, so $\mathbf{C_s = 1\ \mu F}$. In series the charge is the same on each: $Q = C_sV =$ **24 μC**.
2 (OR)	$p = q \times 2l$, the product of either charge and the distance between them, directed from $-q$ to $+q$. SI unit: **coulomb metre** (C m). $\tau = pE \sin\theta$ is maximum when $\boldsymbol{\theta = 90°}$, i.e. the dipole axis perpendicular to the field.
3	$I = 4/(60+40) = 0{\cdot}04$ A; $V_{AB} = (0{\cdot}04)(40) = 1{\cdot}6$ V; $K = 1{\cdot}6/10 =$ $\mathbf{0{\cdot}16\ V\ m^{-1}}$ (= $0{\cdot}016$ V cm^{-1}). Note that only the **wire's** share of the p.d. enters the gradient.
4	Any two: diamagnetics are feebly **repelled** by a magnet, ferromagnetics strongly **attracted**; χ is small and **negative** for a diamagnetic, very large and positive for a ferromagnetic; μ_r is slightly less than 1 against much greater than 1; diamagnetism is independent of temperature while a ferromagnetic becomes paramagnetic above its Curie temperature. Examples: bismuth/copper against iron/cobalt/nickel.
5	Initially $\Phi = NBA$; after the 90° turn the plane contains **B** so $\Phi = 0$. $\Delta\Phi = (100)(0{\cdot}20)(5 \times 10^{-3}) = 0{\cdot}10$ Wb, so $\varepsilon = 0{\cdot}10/0{\cdot}10 =$ **1·0 V**.
5 (OR)	An ideal transformer conserves power: $V_pI_p = V_sI_s$. In a step-down transformer $V_s < V_p$, so I_s must be correspondingly **greater**. It trades voltage for current and creates no energy.
6	(i) **Microwaves**; used in radar, satellite communication or a microwave oven (any one). (ii) Their **speed**, $c = 3 \times 10^8$ m s^{-1}. (Also acceptable: all are transverse, all carry no charge.)
7	(i) The image forms at 2*f* on the **other** side, and is **real, inverted and the same size** as the object ($m = -1$). (ii) The power of a lens is the reciprocal of its focal length in **metres**, $P = 1/f$; SI unit the **dioptre** (D).

Q	Answer
8	(i) β ∝ D, so β = 0·20 × (1·5/1·0) = **0·30 mm**. (ii) β ∝ λ, so a shorter wavelength **decreases** the fringe width – the fringes crowd together.

Section C

Q	Answer
9	Take a cylindrical pill-box Gaussian surface of cross-section A piercing the sheet symmetrically. **[1]** The curved surface is parallel to **E** so contributes no flux; the two flat faces give $\Phi = 2EA$, and the charge enclosed is σA. **[1]** By Gauss' theorem $2EA = \sigma A/\varepsilon_0$, so $\mathbf{E = \sigma/2\varepsilon_0}$, independent of distance. **[1]** (For a **conductor** it is σ/ε_0.)
10	Junction rule: the common arm carries I_1+I_2. Loop 1: $2I_1 + 4(I_1+I_2) = 16$, i.e. $6I_1 + 4I_2 = 16$. Loop 2: $3I_2 + 4(I_1+I_2) = 15$, i.e. $4I_1 + 7I_2 = 15$. Solving, $I_1 = 2$ A, $I_2 = 1$ A, so **I = 3 A**. Check: 2(2)+4(3) = 16 and 3(1)+4(3) = 15.
10 (OR)	**Assumption:** the bridge is balanced, so $I_g = 0$ and B, D are at the same potential. **[1]** Then I_1 flows through both P and Q, and I_2 through both R and S. Loop ABDA: $I_1P = I_2R$. **[1]** Loop BCDB: $I_1Q = I_2S$. Dividing, **P/Q = R/S**. **[1]**
11	Every element dl is at the same distance R from the centre and is **perpendicular** to the line joining it there, so sin θ = 1. **[1]** Hence $dB = (\mu_0/4\pi)I\,dl/R^2$, and every dB points the same way along the axis, so the magnitudes add: $B = (\mu_0 I/4\pi R^2)\oint dl$. **[1]** With $\oint dl = 2\pi R$ per turn and N turns, $\mathbf{B = \mu_0 NI/2R}$, directed along the axis by the right-hand rule. **[1]**
12	(i) $S = I_gG/(I - I_g)$ = (0·01)(99)/(1 − 0·01) = 0·99/0·99 = **1 Ω**, connected as a **shunt in parallel**. (ii) $R_A = GS/(G+S)$ = 99/100 = **0·99 Ω** – smaller than either, as an ammeter must be.
13	$B = \mu_0 nI = \mu_0 NI/l$. **[1]** Flux linkage $N\Phi = N{\cdot}BA = \mu_0 N^2AI/l$. **[1]** Since $N\Phi = LI$, $\mathbf{L = \mu_0 N^2A/l}$. **[1]** It depends on the number of turns (as N^2), the area of cross-section, the length, and the permeability of the core (any two).
14	Diagram: incident plane wavefront AB, A reaching O first, B reaching O′ after time t; i and r marked; refracted wavefronts drawn **closer together** than the incident ones. In that time BO′ = c_1t and OC = c_2t. **[1]** From the right-angled triangles OBO′ and OCO′, sin $i = c_1t$/OO′ and sin $r = c_2t$/OO′. **[1]** Dividing, $\mathbf{\sin i/\sin r = c_1/c_2 = n}$, a constant. **[1]**
14 (OR)	(a) Full angular width = $2\lambda/a = 2(600 \times 10^{-9})/3 \times 10^{-4}$ = $\mathbf{4 \times 10^{-3}}$ **rad**. (b) Linear width = $2\lambda D/a$ = $(4 \times 10^{-3})(1{\cdot}5)$ = **6·0 mm**.
15	Labelled figure with i_1, r_1, r_2, i_2 and δ marked. From the quadrilateral, $\mathbf{A = r_1+r_2}$ and $\mathbf{\delta = i_1+i_2 - A}$. **[1]** At **minimum deviation** the path is symmetric: $i_1 = i_2$ and $r_1 = r_2$, so $r = A/2$ and $i = (A+\delta_m)/2$. **[1]** Snell's law at the first face, $n = \sin i/\sin r$, gives the result. **[1]**
16	**Eyepiece** (v_e = −25 cm): 1/(−25) − 1/u_e = 1/6·25 gives u_e = −5·0 cm, so v_o = 15 − 5 = 10 cm. **Objective**: 1/10 − 1/u_o = 1/2·0 gives u_o = −2·5 cm. Then m_o = 10/2·5 = 4 and m_e = 1 + 25/6·25 = 5, so **M = 20**.
17	(i) The amplitudes are no longer equal, so the minima are **no longer perfectly dark** and the contrast falls; the fringe **width is unchanged**. (ii) A **white central fringe** with a few coloured fringes on either side, which soon overlap into general illumination, because β ∝ λ. (iii) Interference **disappears**; the remaining slit still diffracts, so a **single-slit diffraction pattern** is seen – *not* uniform illumination.

Section D

Q	Answer
18	**(a)** Labelled figure (pole P, centre C, object O, image I, paraxial ray refracted at N). With the small angles α, β, γ at O, I, C: i = α+γ and r = γ − β; Snell for small angles gives $n_1(\alpha+\gamma) = n_2(\gamma-\beta)$; with α ≈ NM/PO, β ≈ NM/PI, γ ≈ NM/PC and the sign convention, $\mathbf{n_2/v - n_1/u = (n_2-n_1)/R}$. [1 figure, 1 angles, 1 result] **(b)** 1·5/v − 1/(−40) = 0·5/15 = 0·03333, so 1·5/v = 0·03333 − 0·025 = 0·008333 and **v = +180 cm** – a real image 180 cm from the pole, inside the glass.
18 (OR)	**(a)** Ray diagram: parallel rays from the distant object, real inverted image at the common focal plane, final image at infinity; f_o, f_e, α and β marked. α ≈ h/f_o and β ≈ h/f_e, so $\mathbf{M = \beta/\alpha = f_o/f_e}$. **(b)** M = 150/6 = **25** and $L = f_o+f_e$ = **156 cm**.

Q	Answer
19	**(a)** $X_L = 2\pi(50)(0{\cdot}50) = 157{\cdot}1\ \Omega$; $Z = \sqrt{(30^2 + 157{\cdot}1^2)} = 159{\cdot}9\ \Omega$; $I_{rms} = 200/159{\cdot}9 =$ **1·25 A**; $\cos\varphi = R/Z = 30/159{\cdot}9 =$ **0·19** (current lagging by about 79°). **(b)** At resonance $\mathbf{X_L = X_C}$, so the net reactance is zero, Z is minimum and equal to R, the current is maximum and in phase with the emf, and $\mathbf{f_0 = 1/2\pi\sqrt{(LC)}}$.
19 (OR)	**(a)** $N_s = 4000 \times (220/2200) =$ **400 turns**. For an ideal transformer $P_p = P_s = 8{\cdot}0$ kW, so $I_p = 8000/2200 =$ **3·6 A**. **(b)** Any two: **eddy-current** loss, reduced by a **laminated** core; **hysteresis** loss, reduced by using **soft iron** with a narrow B–H loop; **copper** (I^2R) loss, reduced by thick low-resistance windings; **flux leakage**, reduced by winding the coils one over the other on a high-permeability core.
20	(i) $U = ½CV^2$ (= $½QV = Q^2/2C$). (ii) $U = ½(100 \times 10^{-6})(300)^2 =$ **4·5 J**. (iii) $\langle P \rangle = 4{\cdot}5/(2{\cdot}0 \times 10^{-3}) =$ **2·25 kW**. (iv) $I_0 = V/R = 300/1{\cdot}0 \times 10^6 = 3 \times 10^{-4}$ A = **0·30 mA**. (v) A capacitor can release its whole stored energy in a very short time, giving a very **high peak power**; the current a battery can supply is limited by its internal resistance, so it could not deliver 2·25 kW even though it stores far more energy in total.

Mark distribution over the First Term chapters

Chapter	A	B	C	D	Total
Electrostatics	2	2	3	3	**10**
Current Electricity	2	2	3	2	**9**
Magnetic Effect of Current and Magnetism	3	2	6	—	**11**
Electromagnetic Induction and Alternating Current	2	2	3	5	**12**
Electromagnetic Waves	1	2	—	—	**3**
Optics — Ray Optics and Optical Instruments	2	2	6	5	**15**
Optics — Wave Optics	2	2	6	—	**10**
Total	**14**	**14**	**27**	**15**	**70**